Project :
Uber Ride Cancellation Prediction

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Objective

- Build a predictive model to determine whether a customer will cancel a ride before it begins, using booking metadata available at the time of booking.

- Goal: Help Uber proactively identify high-risk cancellations and optimize driver dispatch efficiency.

Dataset Overview

- Contains Uber ride booking information (150000 entries) of 2024 of Delhi NCR region including:
  - Booking details (ID, status, datetime)
  - Customer info
  - Vehicle type
  - Pickup/Drop location
  - VTAT (Vehicle Time of Arrival), CTAT (Customer Time of Arrival)
  - Cancellation by Customer/ Driver or Incomplete Ride and Reasons
  - Booking Value & Ride Distance
  - Ratings & Payment info
- Source: Kaggle Uber Ride Analytics Dataset
  https://www.kaggle.com/datasets/yashdevladdha/uber-ride-analytics-dashboard/data

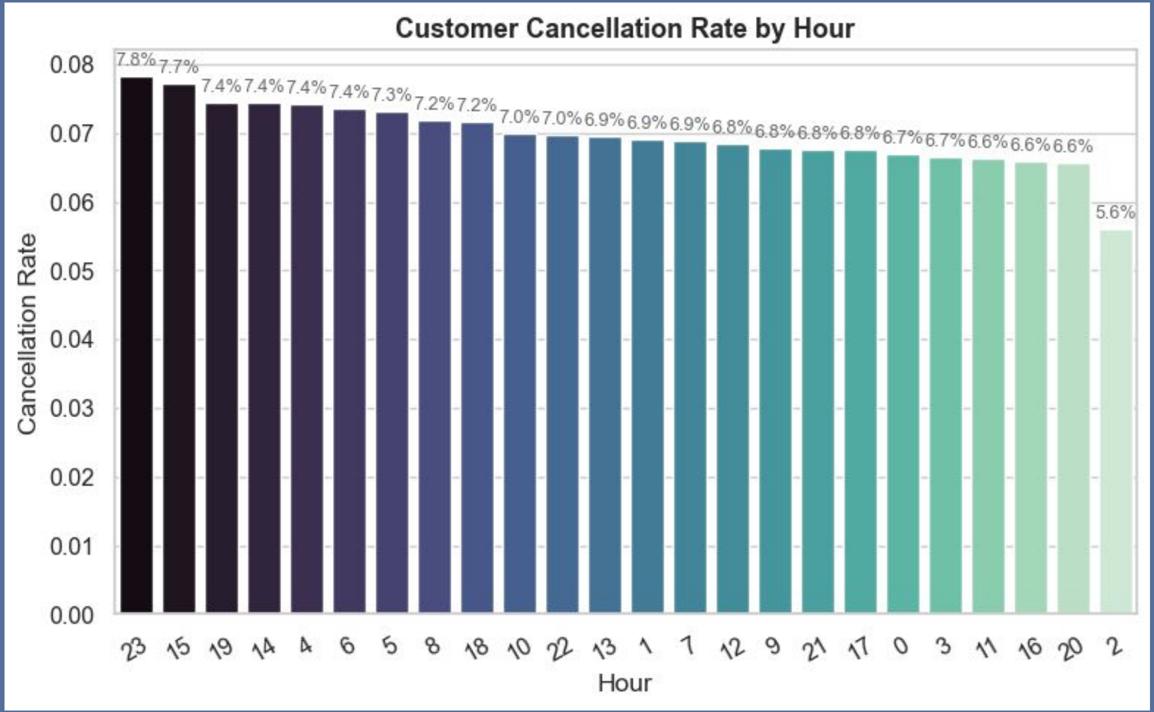
Tech Stack

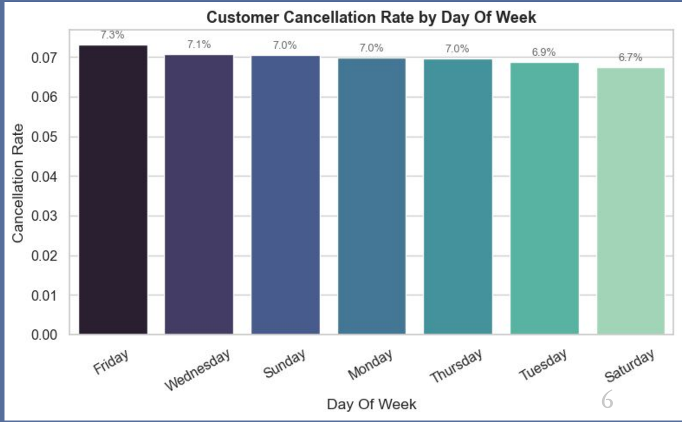
- Python 3.x on VS code
- pandas, numpy – Data manipulation
- matplotlib, seaborn – Visualization
- scikit-learn *(LogisticRegression, RandomForestClassifier, GridSearchCV, MinMaxScaler, metrics)* - Machine learning
- xgboost – Machine learning

Data Preprocessing/ Cleaning

- Standardized column names and values
- Handled null values - median for continuous variables and ‘Unknown’ for categorical.
- Check/eliminate/deal with duplicates or outliers
- Combined date-time features in correct format/ type.
- Derived new features (is_weekend, hour, day, month etc.)

Exploratory Data Analysis





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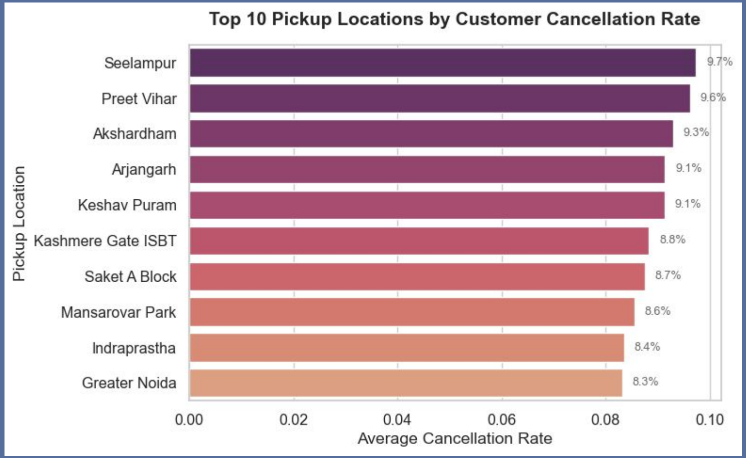

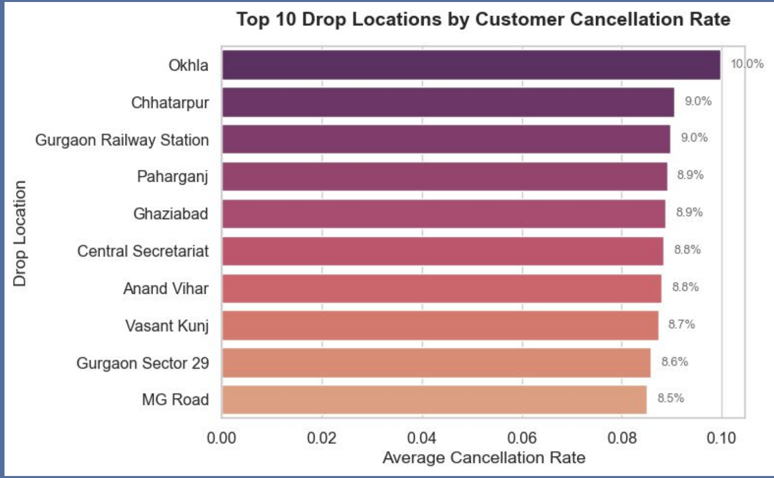

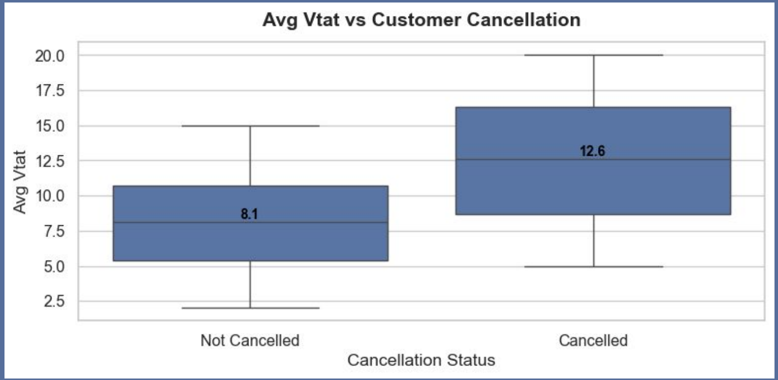

Exploratory Data Analysis

- Key Insights:
  - High VTAT strongly correlates with cancellations.
  - Cancellations peak around 23:00 and 15:00 hours and on Fridays.
  - Specific pickup / drop zones show higher cancellations.

Feature Engineering

- Time-based features (peak hours)
- Location-based encoding (top 10)
  - By Total Occurrence
  - By Cancelled rides
- Customer booking patterns
- One-hot encoding (dummies) for categorical variables
- Dropping unnecessary columns

Model Development

- Three models evaluated:
  - Logistic Regression
  - Random Forest using GridSearchCV
  - XGBoost using GridSearchCV

- Train-Test Split (70/30)
- Feature Scaling - MinMaxScaler()

Model Performance

Model	Accuracy	ROC-AUC	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)
Logistic Regression	0.9402	0.7849	0.989	0.147	0.256
Random Forest ✅ (Best)	**0.9666 ✅**	**0.9538**	**1.000**	**0.523**	**0.690 ✅**
XGBoost	0.956	0.9544	0.741	0.570	0.644

Why F1 score?

- Goal: Predict customer cancellations proactively.

- F1 score chosen to balance:
  - Precision: Accuracy of predicted cancellations.
  - Recall: How many actual cancellations are detected.

- Models optimized for highest F1 to balance detection and reliability.

- Random Forest performed best:
  - F1 score: 69%
  - Precision: 100% → almost no false positives.

- Slightly lower recall than XGBoost, but ensures trustworthy predictions for proactive actions.

Top Features Contributing to Cancellations (Random Forest)

Rank	Feature	Importance
1	**avg_vtat**	**0.6289**
2	booking_value	0.0857
3	ride_distance	0.0855
4	customer_rating	0.0616
5	avg_ctat	0.0451

Business Insights & Impact

Insight	Impact on Cancellations	Business Action / Benefit
High VTAT (Vehicle Arrival Time)	Strong driver – longer waits → higher cancellations	Prioritize rides with long VTAT for proactive actions
Ride distance, booking value, customer ratings	Moderate influence	Consider in risk scoring for cancellations
Random Forest model (F1-focused)	Balanced precision & recall	Enables proactive targeting of high-risk rides, reduces cancellations via driver re-balancing

Recommendations

- Driver rebalancing in high-VTAT zones
- Dynamic dispatching to minimize arrival time
- Customer alerts for high-risk bookings
- Continuous monitoring of VTAT trends

Future Enhancements

- Calculate driver arrival distance
- Integrate real-time traffic and weather data
- Include special event information
- Build live dashboard for cancellation monitoring

Thank You